Autonomous Cave Exploration

Autonomous Systems Project Report

Technical University of Munich | Group 12

March 4, 2026

Mission Overview

Subterranean Navigation

The core objective is to integrate high-level state machines with spatial mapping and trajectory planning for autonomous cave entry and flight.
The drone must build a reliable real-time 3D voxel map while traversing tight, unknown cave spaces

Target Identification

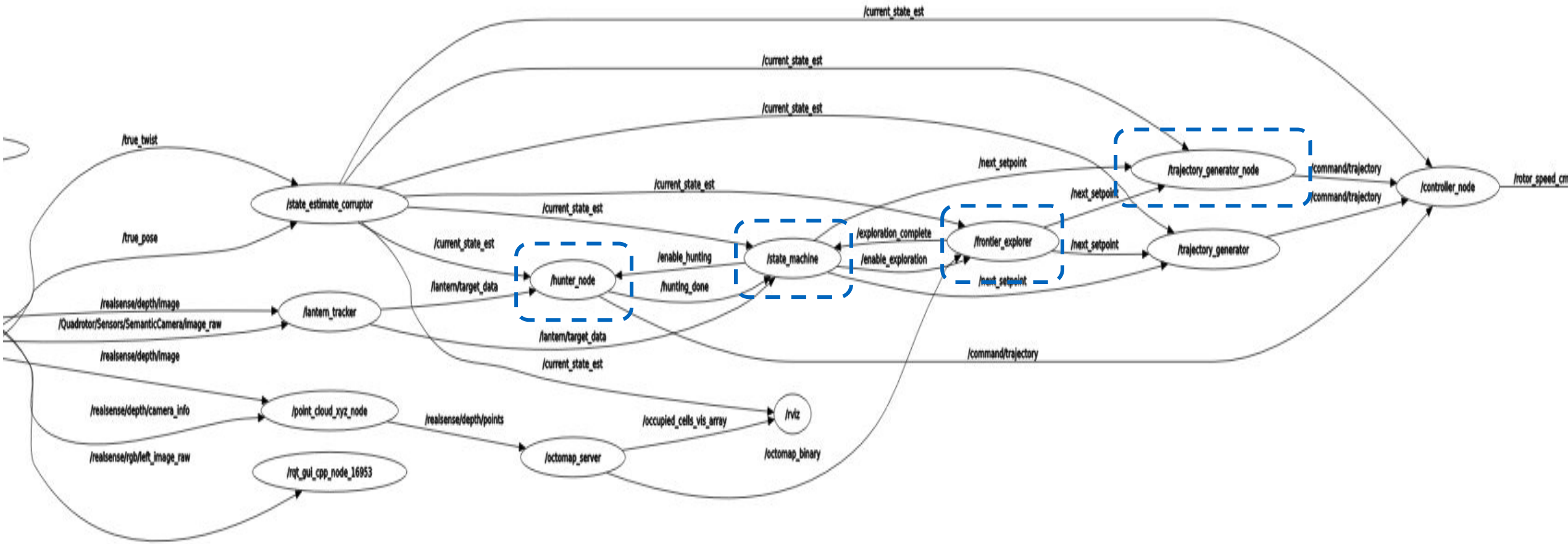
Deploy a semantic perception layer for real-time detection of specific visual targets (lanterns).

Execute precise "hunting" maneuvers to intercept targets, logging spatial coordinates and stabilizing for data capture.

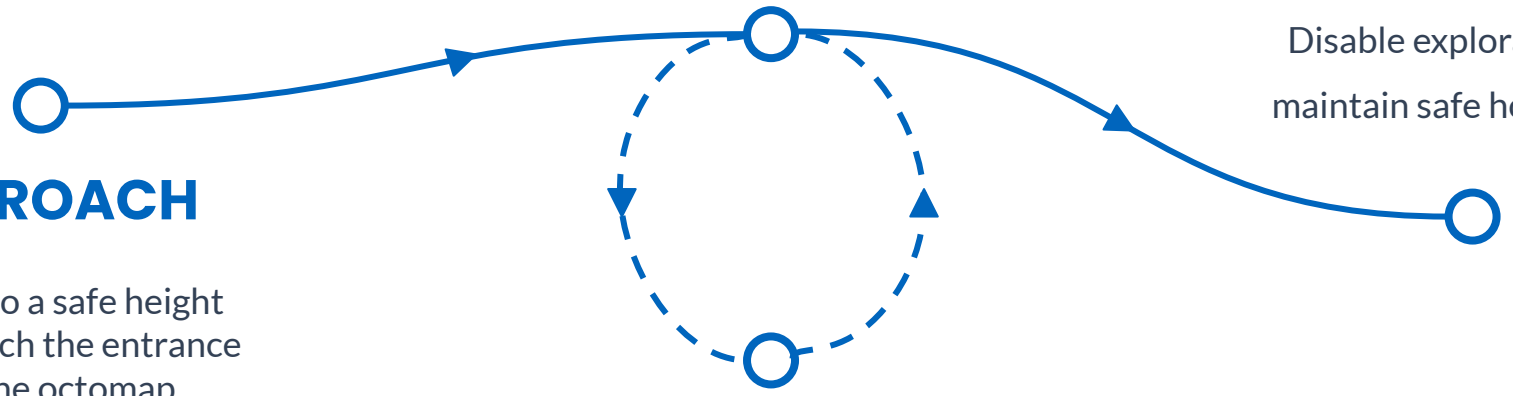
Modular ROS2 Architecture

- **decision_making:** Central State Machine coordinating mission logic and state transitions.
- **exploration_pkg:** Implements frontier-based exploration and backtracking
- **trajectory_generator:** Continuous 10th-order polynomial flight path generation for smooth motion.
- **perception_pkg:** Manages Octomap mapping and semantic lantern detection thresholding.
- **teleop_pkg:** Manual control for testing and debugging.

General Architecture



Mission State Machine – the brain



EXPLORE

Enable graph based frontier exploration algorithm

MISSION COMPLETE

Disable exploration and maintain safe hover state.

APPROACH

- Hover to a safe height
- Approach the entrance
- Reset the octomap
- Enter the cave

LANTERN HUNTING

Enable lantern hunting node

Implemented in branch /mainlanterndetection

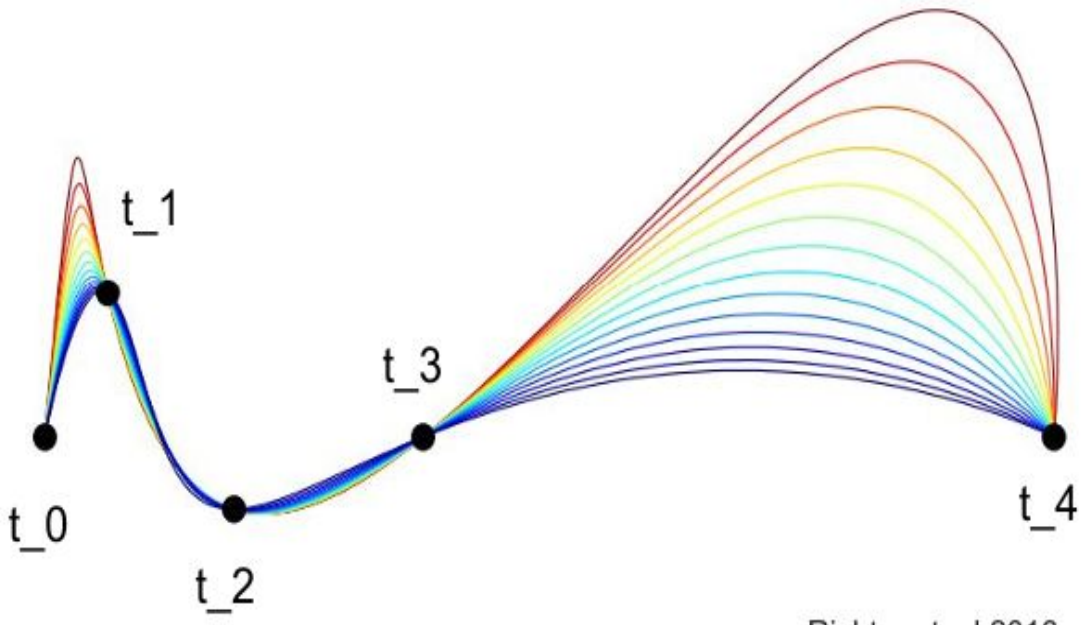
Trajectory Generation

- Dynamic & Reactive Planning: Computes new minimum-acceleration polynomials instantly upon receiving a target, anchoring the start to real-time state data to eliminate positional "jumping."

- Smooth Optimization: Leverages the ETH Zürich MAV trajectory generation framework and a Linear Optimization solver to guarantee continuous, physically feasible flight paths.

- High-Frequency Control: Samples generated trajectories at 100Hz (10ms intervals) to provide a dense, synchronized stream of setpoints for the low-level geometric controller.

- Kinematic Constraints: Safely bounds flight paths with a maximum velocity of 1.0 m/s and a maximum acceleration of 2 m/s^2.



Richter et. al 2016.

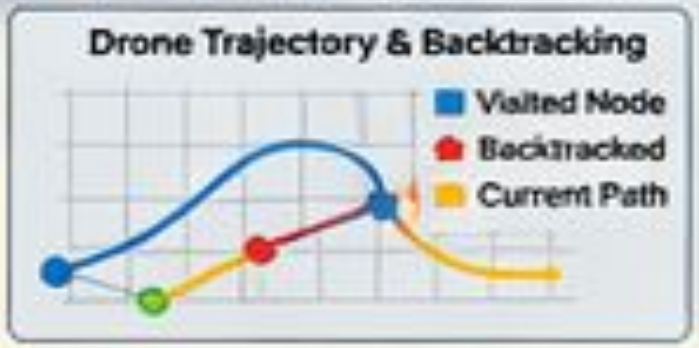
Frontier Explorer

Key Responsibilities:

- Detect frontiers (boundary between known/unknown)
- Cluster & score frontiers (size + alignment bias)
- Select next frontier considering distance & safety
- Handle unreachable goals via backtracking
- Publish exploration setpoints

Algorithm Pipeline:

1. Extract Frontier Voxels
2. Cluster Frontiers
3. Score & Rank
4. Validate (distance/obstacles)
5. Publish Best Goal



Visualization:

- 3D Octomap (explored=blue, unexplored=gray, frontiers=red)
- Heatmap of frontier scores + drone path with backtracking

RViz

Frontier Exploration - Strategy

1. Detection & Clustering

Search Octomap for "Free" voxels adjacent to "Unknown" space within a 60m bounding box.

Clusters are formed using Nearest-Neighbor Centroid logic to find the center of tunnel openings.

2. Heading Bias & Safety

Scoring bias is applied to clusters aligned with current flight direction to minimize oscillations.

Safety node maintains a backtracking chain to ensure a return path is always available if the drone gets stuck.

Frontier Exploration - Backtracking

Negotiation

The exploration is governed by a **negotiation** logic that gradually decrement the minimum cluster size threshold to ensure that the drone doesn?t give un in narrow spaces

Backtracking

When no undiscovered frontiers remain the algorithm switches to backtracking and perform the following:

1. moves back to parent node
2. pops current node from the graph
3. scans for new frontiers

Perception & Mapping Pipeline

Point Cloud

Converts 2D depth imagery into spatial 3D data for obstacle avoidance using depth_image_proc.

Octomap Grid

Builds a voxel-grid representation at 0.6 resolution, reset dynamically prior to cave entry.

Target Hunting

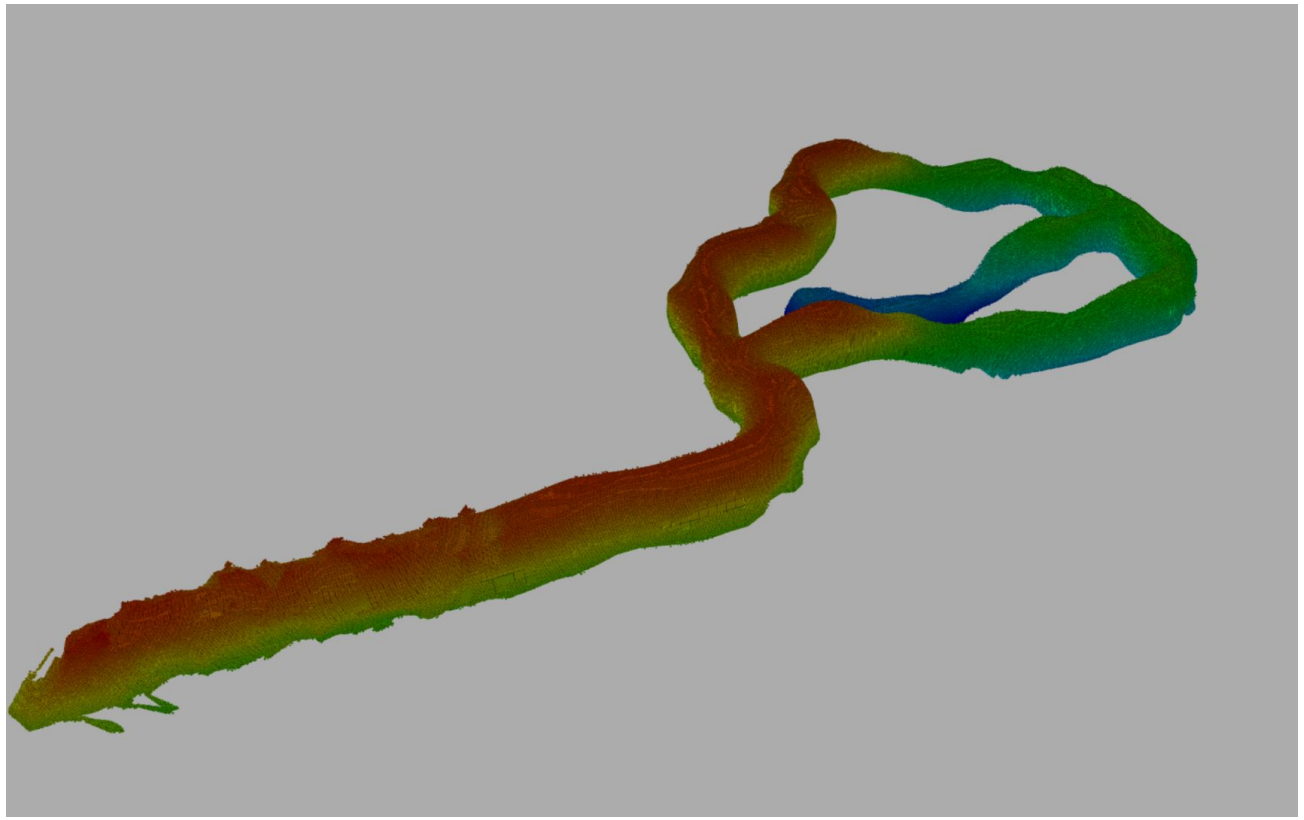
Identifies target via semantic data. Reportes the location of target in the world frame

Generated 3D Cave Map

Voxel-Grid Resolution

The Octomap resolution is set to 0.6 for optimal balance between detail and real-time processing.

- Dynamic mapping allows for immediate frontier detection.
- Pre-entry clearing ensures no artifacts from outside the cave.
- Provides collision-free safety corridors for navigation.



Lantern Hunting strategy

add text here

Lantern Detection Success

Implemented in branch /mainlanterndetection

75%

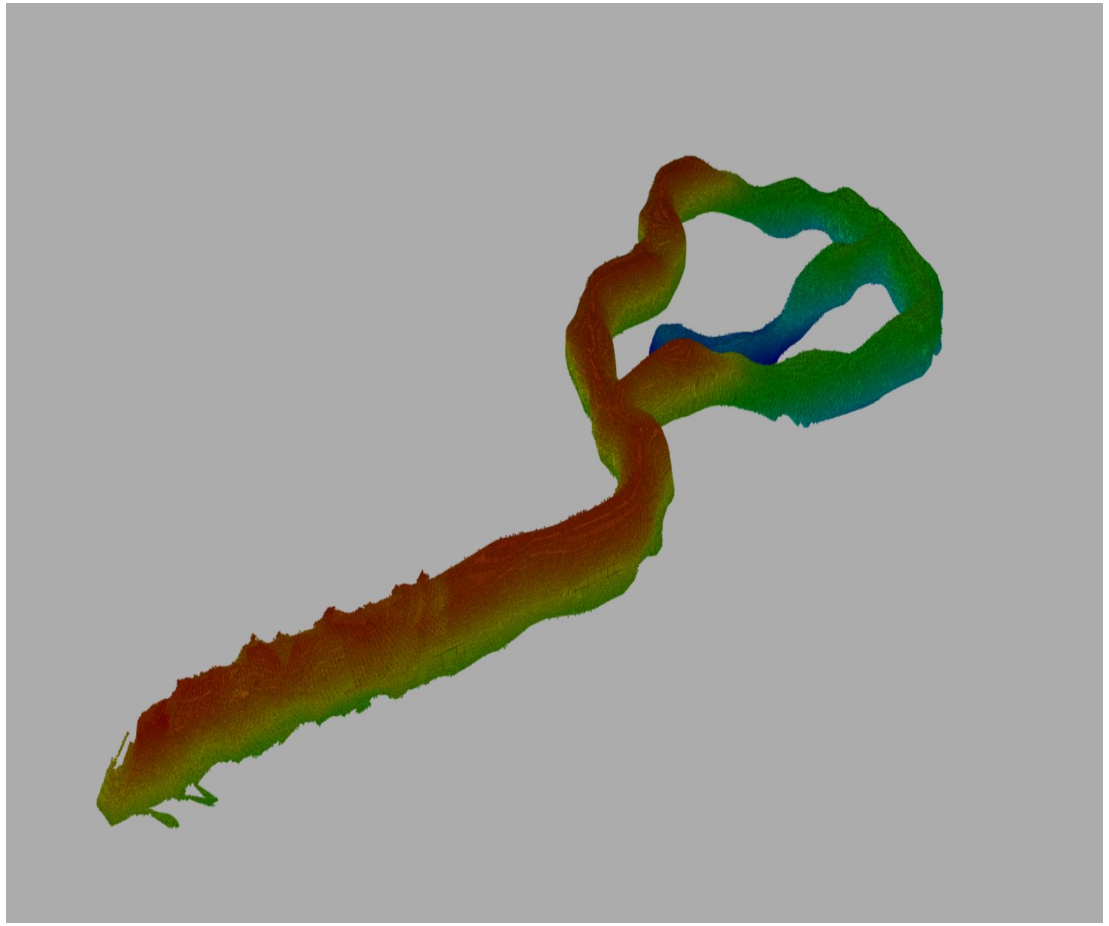
3 / 4 Lanterns Intercepted

Strategy Outcomes

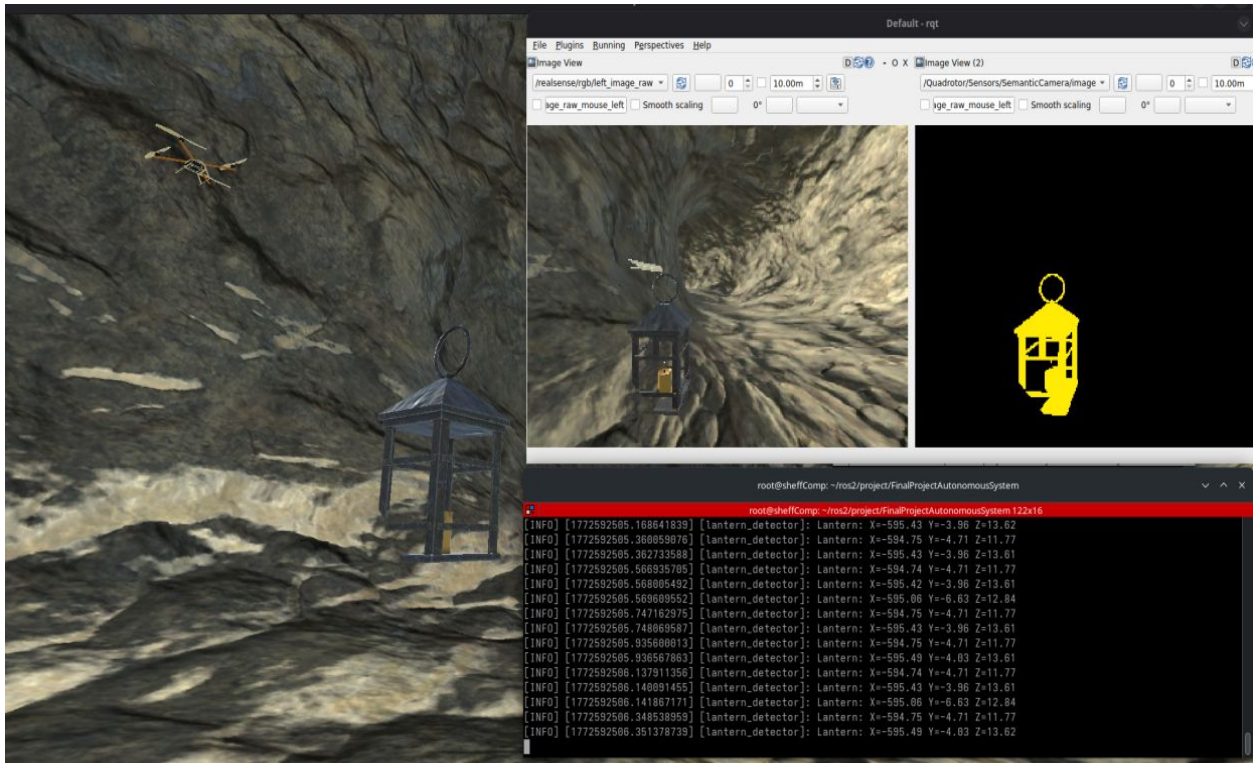
The semantic detection module proved working across varying light conditions inside the cavern.

Missed target analysis: The 4th lantern was located in an area with restricted coverage , preventing a safe path generation within mission time limits.

Results



Complete Cave Exploration in about 35 minutes



Lantern Detection

Team Contributions

Team Member	Primary Responsibilities
Fuzail Hamid Banday	Decision Making Package, Git Management, Testing
Firdavs Shukurilloev	Teleop & Exploration Package (RRT*)
Samuele Ribaudo	Trajectory Generation & Graph-based Exploration
Hong Yan Jun	Perception Package, Lantern Detection Module
Om Anant Kulkarni	Decision Making & Exploration (ERRT, RTAB-Map)

Questions?

Autonomous Systems Project | Group 12

Technical University of Munich
Department of Informatics